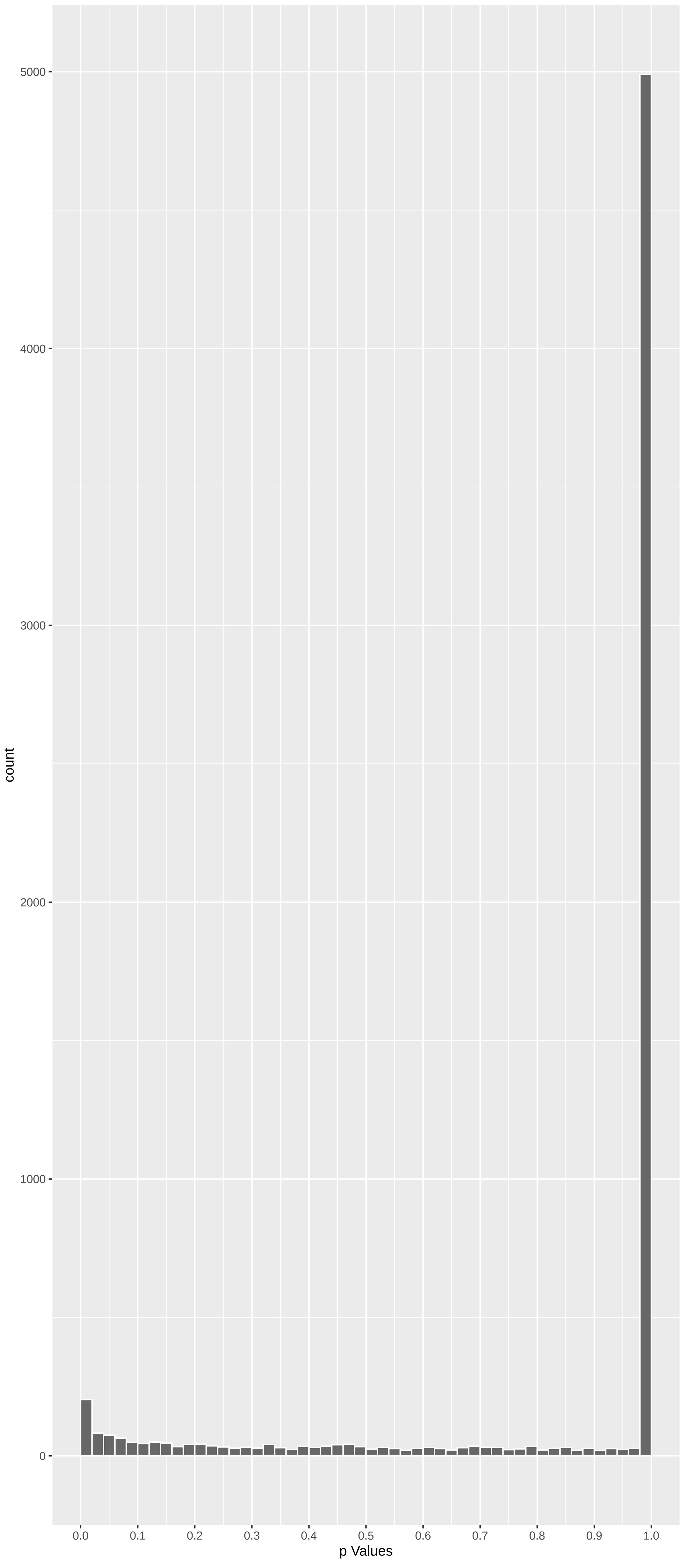
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
ADGRG4	−7.116535	6.640423e−12	1.060e−07	1.693e−03
PCDH11X	−6.490163	5.144621e−10	4.107e−06	3.278e−02
RGPD8	5.889158	2.329007e−08	9.853e−05	2.104e−01
TASOR2	−5.879527	2.468639e−08	9.853e−05	2.104e−01
SUSD5	−5.752469	5.276958e−08	1.076e−04	2.104e−01
GOLGA4	−5.797264	4.044324e−08	1.076e−04	2.104e−01
PIGG	5.748833	5.391681e−08	1.076e−04	2.104e−01
ATP8B2	−5.783592	4.387334e−08	1.076e−04	2.104e−01
HERC2	5.712346	6.685751e−08	1.186e−04	2.104e−01
PRUNE2	−5.666245	8.757651e−08	1.398e−04	2.232e−01
RABEP1	−5.438650	3.221154e−07	4.675e−04	6.785e−01
PER3	−5.352908	5.193099e−07	6.909e−04	8.389e−01
DYNC2H1	−5.314074	6.432067e−07	7.899e−04	8.389e−01
CEP126	−5.297795	7.032565e−07	8.020e−04	8.389e−01
TAS2R42	5.280994	7.709094e−07	8.205e−04	8.389e−01
STAMBPL1	−5.264694	8.425364e−07	8.407e−04	8.389e−01
PCDH11Y	−5.230867	1.012303e−06	9.507e−04	8.928e−01
CAMTA2	−5.138859	1.658467e−06	1.471e−03	1.000e+00
CRTC2	−5.127504	1.761656e−06	1.480e−03	1.000e+00
DHDH	−5.069018	2.399242e−06	1.741e−03	1.000e+00
ENPP2	−5.086120	2.192774e−06	1.741e−03	1.000e+00
LCN12	5.072669	2.353652e−06	1.741e−03	1.000e+00
GPATCH4	−5.051168	2.634702e−06	1.829e−03	1.000e+00
KDM2B	−4.926432	5.024675e−06	3.342e−03	1.000e+00
METTL2B	−4.907404	5.537391e−06	3.536e−03	1.000e+00

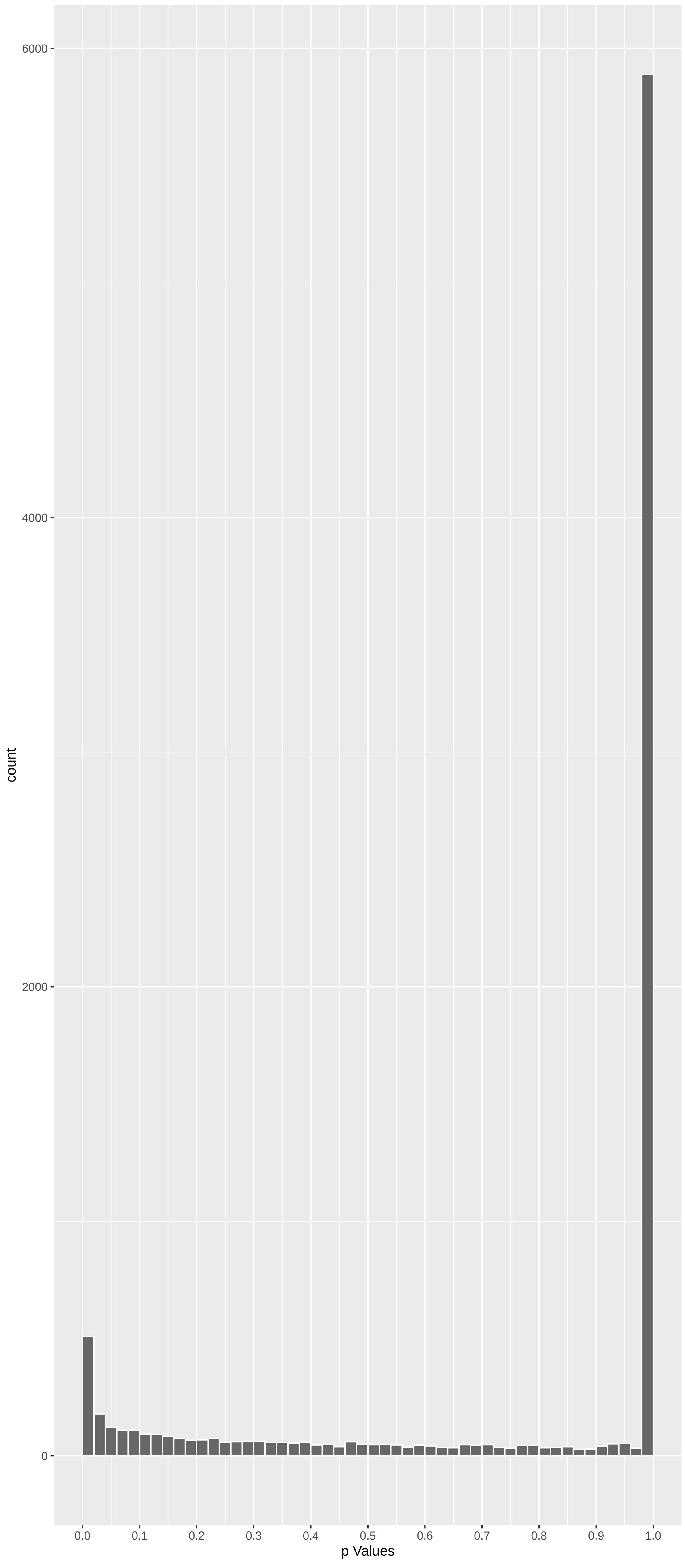
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
ADGRG4	−7.11653546	6.640423e−12	1.060e−07	1.693e−03
PCDH11X	−6.49016278	5.144621e−10	4.107e−06	3.278e−02
HERC2	5.71234608	6.685751e−08	1.186e−04	2.104e−01
RGPD8	5.88915814	2.329007e−08	9.853e−05	2.104e−01
SUSD5	−5.75246918	5.276958e−08	1.076e−04	2.104e−01
TASOR2	−5.87952729	2.468639e−08	9.853e−05	2.104e−01
GOLGA4	−5.79726433	4.044324e−08	1.076e−04	2.104e−01
PIGG	5.74883332	5.391681e−08	1.076e−04	2.104e−01
ATP8B2	−5.78359183	4.387334e−08	1.076e−04	2.104e−01
PRUNE2	−5.66624493	8.757651e−08	1.398e−04	2.232e−01
RABEP1	−5.43864968	3.221154e−07	4.675e−04	6.785e−01
CEP126	−5.29779506	7.032565e−07	8.020e−04	8.389e−01
TAS2R42	5.28099401	7.709094e−07	8.205e−04	8.389e−01
STAMBPL1	−5.26469442	8.425364e−07	8.407e−04	8.389e−01
PER3	−5.35290843	5.193099e−07	6.909e−04	8.389e−01
DYNC2H1	−5.31407389	6.432067e−07	7.899e−04	8.389e−01
PCDH11Y	−5.23086661	1.012303e−06	9.507e−04	8.928e−01
M6PR	−1.23304919	1.000000e+00	1.000e+00	1.000e+00
FKBP4	0.87499057	1.000000e+00	1.000e+00	1.000e+00
CYP26B1	−2.32950982	1.189924e−01	1.000e+00	1.000e+00
NDUFAF7	−1.56023899	7.122205e−01	1.000e+00	1.000e+00
FUCA2	−1.12324329	1.000000e+00	1.000e+00	1.000e+00
DBNDD1	−0.09622794	1.000000e+00	1.000e+00	1.000e+00
HS3ST1	1.40230077	9.649527e−01	1.000e+00	1.000e+00
SEMA3F	−0.06864406	1.000000e+00	1.000e+00	1.000e+00

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
RGPD8	5.889158	2.329007e−08	9.853e−05	2.104e−01
PIGG	5.748833	5.391681e−08	1.076e−04	2.104e−01
HERC2	5.712346	6.685751e−08	1.186e−04	2.104e−01
TAS2R42	5.280994	7.709094e−07	8.205e−04	8.389e−01
LCN12	5.072669	2.353652e−06	1.741e−03	1.000e+00
GRK7	4.863147	6.932047e−06	3.816e−03	1.000e+00
PDE6B	4.659562	1.901295e−05	7.589e−03	1.000e+00
CLU	4.652043	1.971971e−05	7.679e−03	1.000e+00
ENTPD1	4.594433	2.603561e−05	8.607e−03	1.000e+00
OR6N2	4.593046	2.620942e−05	8.607e−03	1.000e+00

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
ADGRG4	−7.116535	6.640423e−12	1.060e−07	1.693e−03
PCDH11X	−6.490163	5.144621e−10	4.107e−06	3.278e−02
TASOR2	−5.879527	2.468639e−08	9.853e−05	2.104e−01
SUSD5	−5.752469	5.276958e−08	1.076e−04	2.104e−01
GOLGA4	−5.797264	4.044324e−08	1.076e−04	2.104e−01
ATP8B2	−5.783592	4.387334e−08	1.076e−04	2.104e−01
PRUNE2	−5.666245	8.757651e−08	1.398e−04	2.232e−01
RABEP1	−5.438650	3.221154e−07	4.675e−04	6.785e−01
PER3	−5.352908	5.193099e−07	6.909e−04	8.389e−01
DYNC2H1	−5.314074	6.432067e−07	7.899e−04	8.389e−01

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Fatty Acid Beta−Oxidation (GO:0006635)	−0.17561040	47	3.148e−05	1.628e−01	CPT2:128 ACAT1:276 ACADM:330 ACAD10:372 ACAD11:525 ACADVL:663
Fatty Acid Oxidation (GO:0019395)	−0.16922382	47	6.044e−05	1.628e−01	CPT2:128 PHYH:182 ACAT1:276 ACADM:330 ACAD10:372 ACAD11:525
DNA Recombination (GO:0006310)	−0.18073814	39	9.474e−05	1.702e−01	RAG1:184 RAD50:211 HMGB3:229 BRCA1:377 MSH3:766 DMC1:769
Double−Strand Break Repair (GO:0006302)	−0.08834218	157	1.389e−04	1.871e−01	EME2:80 EME1:88 RAD50:211 BRCA2:230 MUS81:240 ATM:252
Mitochondrial RNA Processing (GO:0000963	−0.35492413	9	2.269e−04	2.445e−01	FASTKD5:287 FASTKD3:441 TRMT10C:1366 PNPT1:1492 TBRG4:1680 SUPV3L1:2101
Regulation Of Mitochondrial mRNA Stabili	−0.41491214	6	4.321e−04	3.880e−01	FASTKD5:287 FASTKD3:441 PDE12:882 TBRG4:1680 FASTKD1:2463 FASTKD2:2528
DNA Repair (GO:0006281)	−0.05913871	272	8.335e−04	4.182e−01	USP45:60 EME2:80 EME1:88 MLH1:199 RAD50:211 BRCA2:230
Calcium Ion Import Across Plasma Membran	0.16248862	33	1.242e−03	4.182e−01	CLU:8 CACNA1B:28 TRPM1:50 CACNA1S:84 TRPV1:154 SCN9A:299
Double−Strand Break Repair Via Homologou	−0.09213794	105	1.127e−03	4.182e−01	BRCA2:230 MUS81:240 ERCC5:357 BRCA1:377 STON1:457 ZMYND8:505
Fatty Acid Catabolic Process (GO:0009062	−0.12776179	58	7.721e−04	4.182e−01	CPT2:128 PHYH:182 ACAT1:276 ACADM:330 ACAD10:372 ACAD11:525
Mitochondrial Gene Expression (GO:014005	−0.09461700	100	1.097e−03	4.182e−01	PTCD3:243 HARS1:248 FASTKD5:287 FASTKD3:441 LARS2:523 MRPS18A:526
Modulation Of Excitatory Postsynaptic Po	0.18068816	29	7.609e−04	4.182e−01	TMEM108:18 NLGN4X:100 CHRNA7:211 PTK2B:417 SHANK3:507 LRRK2:992
Peptide Cross−Linking (GO:0018149)	0.22749467	17	1.167e−03	4.182e−01	FN1:450 F13B:452 COL3A1:781 FGFR1:977 DSP:1269 CSTA:1371
Recombinational Repair (GO:0000725)	−0.10112314	89	9.921e−04	4.182e−01	BRCA2:230 ERCC5:357 BRCA1:377 STON1:457 ZMYND8:505 SLX4:568
Regulation Of Superoxide Anion Generatio	0.28181782	12	7.247e−04	4.182e−01	TGFB1:55 CD177:361 F2RL1:407 TYROBP:600 SOD1:1119 ITGAM:1304
Synaptic Transmission, GABAergic (GO:005	0.25229173	14	1.083e−03	4.182e−01	GABRB3:27 GABRA5:215 GABRG3:264 HAPLN4:693 GABBR2:709 GABRG1:845
Resolution Of Meiotic Recombination Inte	−0.24712241	14	1.369e−03	4.338e−01	EME2:80 EME1:88 MUS81:240 SLX4:568 RMI1:1204 FANCM:1910
DNA Metabolic Process (GO:0006259)	−0.05346433	271	2.567e−03	4.609e−01	USP45:60 RAG1:184 RAD50:211 HMGB3:229 MUS81:240 ATM:252
Amylin Receptor Signaling Pathway (GO:00	0.45090792	4	1.788e−03	4.609e−01	CALCR:233 RAMP3:486 RAMP2:760 RAMP1:1627 NA NA
Calcium Ion Transmembrane Import Into Cy	0.10006760	77	2.427e−03	4.609e−01	CLU:8 CACNA1B:28 TRPM1:50 CACNA1S:84 TRPV1:154 RYR2:254
Histone Ubiquitination (GO:0016574)	0.33179298	7	2.366e−03	4.609e−01	UBR2:96 RNF20:238 UBE2E1:1071 UBE2N:1171 RNF40:1596 WAC:7257
Mitochondrial Translation (GO:0032543)	−0.09288237	96	1.688e−03	4.609e−01	PTCD3:243 HARS1:248 MTIF2:389 LARS2:523 MRPS18A:526 MRPS23:586
Modulation Of Chemical Synaptic Transmis	0.08144676	117	2.387e−03	4.609e−01	TMEM108:18 CACNA1B:28 NLGN4X:100 SLC6A9:176 CHRNA7:211 GRM5:273
Negative Regulation Of Actin Filament Bu	0.17430825	25	2.562e−03	4.609e−01	PRKN:89 TJP1:112 CLASP1:237 SHANK3:507 ARHGAP6:535 MET:1130
Positive Regulation Of Superoxide Anion	0.26396086	11	2.436e−03	4.609e−01	TGFB1:55 F2RL1:407 TYROBP:600 SOD1:1119 ITGAM:1304 FPR2:1464
Positive Regulation Of Transcription By	0.25789918	12	1.980e−03	4.609e−01	DDX21:214 SMARCA5:284 RPTOR:408 CHD8:459 DEK:923 ZC3H8:949
Positive Regulation Of Tyrosine Phosphor	0.12504011	49	2.478e−03	4.609e−01	CSH2:167 IL6ST:206 LIF:225 CD40:293 IL31RA:331 GH1:394
Ribonucleoprotein Complex Biogenesis (GO	−0.09082664	95	2.252e−03	4.609e−01	NOL7:31 DNTTIP2:338 RRP36:431 GRWD1:624 DDX31:772 UTP3:855
Ribosomal Small Subunit Biogenesis (GO:0	−0.11465874	63	1.664e−03	4.609e−01	NOL7:31 DNTTIP2:338 RRP36:431 UTP3:855 TBL3:929 RIOK2:978
Ribosome Biogenesis (GO:0042254)	−0.07725985	129	2.496e−03	4.609e−01	NOL7:31 DNTTIP2:338 RRP36:431 GRWD1:624 DDX31:772 UTP3:855
Nucleic Acid Phosphodiester Bond Hydroly	−0.18908352	21	2.709e−03	4.709e−01	RAD50:211 ENPP1:641 PDE12:882 DNA2:1065 ANKLE1:1433 EXD2:1617
Mitochondrial RNA Metabolic Process (GO:	−0.19720012	19	2.927e−03	4.929e−01	FASTKD5:287 FASTKD3:441 MRPL12:934 POLRMT:1275 TFB2M:1365 PNPT1:1492
Fatty Acid Beta−Oxidation Using acyl−CoA	−0.26651929	10	3.520e−03	5.125e−01	ACADM:330 ACAD11:525 ACADVL:663 ETFB:1702 ACADL:2149 ACADS:2859
Inorganic Cation Transmembrane Transport	0.05119797	277	3.516e−03	5.125e−01	CLU:8 KCNK10:17 CACNA2D4:38 TRPM1:50 CACNA1S:84 ABCC9:91
Maturation Of SSU−rRNA From Tricistronic	−0.16892555	25	3.469e−03	5.125e−01	TSR1:300 RRP36:431 UTP3:855 TBL3:929 TSR2:1180 KRI1:1368
Mismatch Repair (GO:0006298)	−0.17739949	23	3.236e−03	5.125e−01	MLH1:199 MLH3:328 MUTYH:448 PMS2:732 MSH3:766 RPA1:903
tRNA Methylation (GO:0030488)	−0.14216924	36	3.172e−03	5.125e−01	METTL2B:20 METTL8:241 NSUN6:296 TRMT1:339 METTL1:428 THADA:1119
Base−Excision Repair (GO:0006284)	−0.13612543	36	4.726e−03	5.788e−01	ERCC5:357 MUTYH:448 OGG1:809 RPA1:903 DNA2:1065 LIG1:1420
Calcitonin Family Receptor Signaling Pat	0.36934512	5	4.233e−03	5.788e−01	CALCR:233 RAMP3:486 RAMP2:760 RAMP1:1627 CALCRL:7257 NA
Monoatomic Cation Transmembrane Transpor	0.05039179	273	4.340e−03	5.788e−01	CLU:8 KCNK10:17 CACNA2D4:38 TRPM1:50 CNGB3:52 CACNA1S:84

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	−0.29669980	33	3.684e−09	2.390e−05	ADAM15:35 PBXIP1:157 FLAD1:270 RUSC1:346 PYGO2:352 THBS3:381
REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN	−0.16720361	63	4.477e−06	1.453e−02	EME2:80 EME1:88 RAD50:211 BRCA2:230 MUS81:240 ATM:252
REACTOME_DISEASES_OF_DNA_REPAIR	−0.18396382	50	6.834e−06	1.478e−02	MLH1:199 RAD50:211 BRCA2:230 ATM:252 BRCA1:377 MUTYH:448
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.05244743	599	1.300e−05	1.999e−02	IL1RL2:53 TGFB1:55 NUP210:64 NUP133:93 RNASEL:151 IL36B:163
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	−0.10949397	131	1.540e−05	1.999e−02	EME2:80 EME1:88 RAD50:211 BRCA2:230 MUS81:240 ATM:252
TURASHVILI_BREAST_LOBULAR_CARCINOMA_VS_L	0.14773868	67	2.916e−05	3.153e−02	COL6A3:35 PDE8A:184 FBN1:213 ADAM12:232 CDK8:240 COL1A2:297
REACTOME_HDR_THROUGH_SINGLE_STRAND_ANNEA	−0.19755687	36	4.109e−05	3.590e−02	RAD50:211 ATM:252 BRCA1:377 ABL1:765 RPA1:903 DNA2:1065
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	−0.11034843	115	4.427e−05	3.590e−02	MLH1:199 RAD50:211 BRCA2:230 ATM:252 FANCE:261 POLH:356
REACTOME_HOMOLOGY_DIRECTED_REPAIR	−0.10919069	103	1.303e−04	8.451e−02	EME2:80 EME1:88 RAD50:211 BRCA2:230 MUS81:240 ATM:252
KAUFFMANN_DNA_REPAIR_GENES	−0.07685212	210	1.272e−04	8.451e−02	EME1:88 MLH1:199 RAD50:211 BRCA2:230 MUS81:240 ATM:252
BRIDEAU_IMPRINTED_GENES	0.15356402	50	1.731e−04	9.659e−02	ATP10A:22 CNTN3:67 UBE3A:87 CD81:170 IGF2R:197 IGF2:202
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	−0.18740622	33	1.951e−04	9.659e−02	EME2:80 EME1:88 RAD50:211 BRCA2:230 MUS81:240 ATM:252
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.04838640	506	2.084e−04	9.659e−02	CHL1:45 SLC20A1:56 CHGB:82 SEC23A:108 SLC25A12:121 SYNJ1:174
BROWNE_HCMV_INFECTION_6HR_UP	0.15116609	51	1.889e−04	9.659e−02	PTGDS:319 EIF2AK2:568 ISG15:603 MX1:718 SFPQ:729 ASCL1:743
REACTOME_NEURONAL_SYSTEM	0.05349908	382	3.471e−04	1.501e−01	KCNK10:17 GABRB3:27 CACNA1B:28 ABCC9:91 GLUL:92 NLGN4X:100
REACTOME_HOMOLOGOUS_DNA_PAIRING_AND_STRA	−0.15661874	42	4.460e−04	1.702e−01	RAD50:211 BRCA2:230 ATM:252 BRCA1:377 RPA1:903 DNA2:1065
TURASHVILI_BREAST_LOBULAR_CARCINOMA_VS_D	0.12846199	63	4.237e−04	1.702e−01	COL6A3:35 FBN1:213 ADAM12:232 COL1A2:297 SMCHD1:318 COL11A1:338
WP_ALLOGRAFT_REJECTION	0.12046671	70	4.949e−04	1.784e−01	TGFB1:55 CD86:181 ABCB1:235 CD80:244 CD40:293 CD28:357
ZHONG_SECRETOME_OF_LUNG_CANCER_AND_FIBRO	0.09310226	113	6.359e−04	2.033e−01	CLU:8 CD9:54 KXD1:60 CD81:170 VCAM1:375 FN1:450
WP_SARSCOV2_INNATE_IMMUNITY_EVASION_AND_	0.13405743	54	6.580e−04	2.033e−01	TGFB1:55 TRAF2:171 CXCR2:189 CXCL13:416 IFNAR1:671 TNF:688
PID_MYC_ACTIV_PATHWAY	0.11500863	74	6.287e−04	2.033e−01	CAD:58 CDCA7:216 NCL:285 TAF4B:351 CDK4:500 MYC:637
KEGG_NON_HOMOLOGOUS_END_JOINING	−0.26785732	13	8.257e−04	2.061e−01	RAD50:211 XRCC5:838 DNTT:1001 POLL:1452 NHEJ1:1511 XRCC4:1915
REACTOME_INTEGRIN_CELL_SURFACE_INTERACTI	0.10591061	83	8.577e−04	2.061e−01	COL7A1:13 COL6A3:35 FBN1:213 COL1A2:297 COL4A2:340 VCAM1:375
REACTOME_DNA_REPAIR	−0.05881031	278	7.620e−04	2.061e−01	USP45:60 EME2:80 EME1:88 MLH1:199 RAD50:211 BRCA2:230
REACTOME_INTERFERON_SIGNALING	0.08162780	141	8.308e−04	2.061e−01	NUP210:64 NUP133:93 RNASEL:151 ABCE1:217 SAMHD1:310 IFIT1:368
WP_DISRUPTION_OF_POSTSYNAPTIC_SIGNALING_	0.17116665	32	8.063e−04	2.061e−01	NLGN4X:100 TJP1:112 RYR2:254 NRXN2:701 CAMK2D:874 DLG1:892
KEGG_ALLOGRAFT_REJECTION	0.20542761	22	8.520e−04	2.061e−01	CD86:181 CD80:244 CD40:293 CD28:357 TNF:688 IL4:1308
REACTOME_FANCONI_ANEMIA_PATHWAY	−0.16214947	35	9.026e−04	2.092e−01	EME2:80 EME1:88 MUS81:240 FANCE:261 DCLRE1A:294 SLX4:568
MIKKELSEN_ES_ICP_WITH_H3K4ME3	−0.03886566	629	9.408e−04	2.105e−01	SLC2A10:30 SPHKAP:55 ALKBH8:57 ZBTB4:64 ANXA3:91 RAB13:113
PID_INTEGRIN1_PATHWAY	0.11779621	65	1.027e−03	2.222e−01	COL7A1:13 COL6A3:35 CD81:170 FBN1:213 COL1A2:297 COL11A1:338
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.04040656	550	1.264e−03	2.278e−01	HRH2:32 CHGB:82 PYY:113 TMEM130:114 PCNX2:123 MAL:207
REACTOME_IMMUNOREGULATORY_INTERACTIONS_B	0.11485063	67	1.156e−03	2.278e−01	CD81:170 CD40:293 COL1A2:297 NCR1:374 VCAM1:375 ICAM1:479
REACTOME_ION_HOMEOSTASIS	0.13759371	46	1.247e−03	2.278e−01	ABCC9:91 RYR2:254 ATP1A3:359 ATP2A3:598 ATP2B1:617 ATP2B3:770
WP_NEUROINFLAMMATION_AND_GLUTAMATERGIC_S	0.08075121	136	1.163e−03	2.278e−01	TGFB1:55 GLUL:92 SLC6A9:176 IL6ST:206 LIF:225 GRM5:273
BIOCARTA_TH1TH2_PATHWAY	0.21611445	19	1.110e−03	2.278e−01	CD86:181 IL18R1:205 CD40:293 CD28:357 IL18:1140 IL4:1308
KEGG_CELL_ADHESION_MOLECULES_CAMS	0.08761921	114	1.243e−03	2.278e−01	NLGN4X:100 CD86:181 L1CAM:188 CLDN8:222 CD80:244 CD40:293
JIANG_AGING_CEREBRAL_CORTEX_DN	0.14421975	41	1.400e−03	2.390e−01	NCL:285 PTGDS:319 DNM1:488 ACO2:531 DYNC1H1:679 NME1:889
KEGG_MISMATCH_REPAIR	−0.20173377	21	1.373e−03	2.390e−01	MLH1:199 MLH3:328 POLD2:547 PMS2:732 MSH3:766 RPA1:903
MIKHAYLOVA_OXIDATIVE_STRESS_RESPONSE_VIA	0.36805868	6	1.794e−03	2.476e−01	OAT:280 HSPB1:925 CALU:1227 PGAM1:1396 AKR1B1:1549 CTSD:7257
BIOCARTA_WNT_LRP6_PATHWAY	−0.37218335	6	1.592e−03	2.476e−01	WNT8B:599 KREMEN2:1305 DKK1:1317 DKK2:2971 WNT8A:3060 FZD1:3098

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Schizophrenia	0.03810082	1604	7.737e−07	7.590e−03	HERC2:3 CLU:8 NAV1:19 APOA1:23 GABRB3:27 CACNA1B:28
Alzheimer's Disease	0.03593076	1638	2.572e−06	1.028e−02	CLU:8 PROM1:15 APOA1:23 BCAM:40 CHL1:45 TGFB1:55
HIV Infections	0.05536350	621	3.145e−06	1.028e−02	ENTPD1:9 PROM1:15 APOA1:23 TLR8:43 CD9:54 TGFB1:55
Aortic Aneurysm, Abdominal	0.08504822	228	1.041e−05	1.702e−02	APOA1:23 TGFB1:55 CAD:58 CNTN3:67 ACE:86 MMP14:111
Hepatitis C	0.05448550	589	7.700e−06	1.702e−02	CLU:8 APOA1:23 TLR8:43 TGFB1:55 TNFSF10:72 ACE:86
Virus Diseases	0.05067483	672	9.477e−06	1.702e−02	PROM1:15 TLR8:43 CD9:54 TGFB1:55 TNFSF10:72 PRKN:89
Autism Spectrum Disorders	0.05576311	480	3.303e−05	4.629e−02	HERC2:3 ATP10A:22 GABRB3:27 UBE3A:87 PRKN:89 GLUL:92
Psoriasis	0.04791714	632	4.743e−05	5.170e−02	APOA1:23 KRT5:36 TLR8:43 IL1RL2:53 TGFB1:55 KLK7:68
Seminoma	0.09089241	171	4.324e−05	5.170e−02	CLU:8 PROM1:15 ACE:86 CSH2:167 IGF2:202 GGTLC3:219
Central neuroblastoma	0.03303334	1383	5.875e−05	5.574e−02	HERC2:3 CLU:8 PROM1:15 CHL1:45 CD9:54 TGFB1:55
Inflammatory dermatosis	0.10149342	131	6.250e−05	5.574e−02	TGFB1:55 KLK8:94 PPARA:155 IL33:172 CXCR2:189 CD40:293
Amyloidosis	0.04415002	699	8.463e−05	6.919e−02	CLU:8 APOA1:23 TGFB1:55 GJA8:59 ANO5:62 ACE:86
Hepatitis B	0.04582506	638	9.314e−05	7.029e−02	PROM1:15 APOA1:23 TLR8:43 SIAH1:48 CD9:54 TGFB1:55
Dermatitis	0.08892047	156	1.316e−04	9.161e−02	TGFB1:55 KLK7:68 KLK8:94 PPARA:155 IL33:172 CXCR2:189
Neuroblastoma	0.03088272	1422	1.434e−04	9.161e−02	HERC2:3 CLU:8 PROM1:15 CHL1:45 CD9:54 TGFB1:55
Ulcerative Colitis	0.04236294	707	1.494e−04	9.161e−02	HERC2:3 CLU:8 PROM1:15 KIF21B:21 TGFB1:55 ACE:86
Presenile dementia	0.06224094	307	1.901e−04	1.097e−01	CLU:8 APOA1:23 ACE:86 PRKN:89 PYY:113 PPARA:155
Crohn Disease	0.04166947	701	2.032e−04	1.107e−01	HERC2:3 ENTPD1:9 COL7A1:13 KIF21B:21 APOA1:23 TLR8:43
Autistic Disorder	0.04432527	608	2.203e−04	1.112e−01	ATP10A:22 GABRB3:27 CHL1:45 GJA8:59 CNTN3:67 ACE:86
Herpes Simplex Infections	0.05340007	405	2.493e−04	1.112e−01	PROM1:15 TGFB1:55 PRKN:89 RCOR1:146 TRPV1:154 CSH2:167
Inflammatory Bowel Diseases	0.04082281	711	2.484e−04	1.112e−01	ENTPD1:9 KIF21B:21 TLR8:43 TGFB1:55 ACE:86 MMP14:111
Pervasive Development Disorder	0.09997659	113	2.473e−04	1.112e−01	HERC2:3 ATP10A:22 PRKN:89 NLGN4X:100 SLC25A12:121 EHMT1:252
Epilepsy, Temporal Lobe	0.08236800	161	3.215e−04	1.315e−01	GABRB3:27 TNFSF10:72 ACE:86 GLUL:92 DEPDC5:152 TRPV1:154
Mental disorders	0.06269432	280	3.259e−04	1.315e−01	KCNK10:17 CACNA2D4:38 ACE:86 KLK8:94 MYT1:179 PI4KA:201
Pain	0.05209958	408	3.351e−04	1.315e−01	TGFB1:55 ACE:86 CLTCL1:150 TRPV1:154 PPARA:155 IL33:172
Cervical Squamous Intraepithelial Neopla	0.15759486	43	3.520e−04	1.328e−01	TGFB1:55 MAL:207 EGFR:373 FN1:450 IL7:514 MYC:637
Hyperactive behavior	0.04228525	622	3.669e−04	1.333e−01	HERC2:3 PROM1:15 GABRB3:27 AKR1D1:29 TGFB1:55 ANO5:62
Chronic Lymphocytic Leukemia	0.03594296	863	4.132e−04	1.361e−01	ENTPD1:9 PROM1:15 ABCC11:16 CD9:54 TGFB1:55 TNFSF10:72
Encephalomyelitis	0.06006599	295	4.124e−04	1.361e−01	CLU:8 ENTPD1:9 APOA1:23 HRH2:32 TLR8:43 TGFB1:55
Hemiplegia	0.23393376	19	4.163e−04	1.361e−01	COL4A2:340 ATP1A3:359 SCN1A:428 COL4A1:461 TNF:688 SLC1A3:1034
Hepatitis	0.06467389	243	5.426e−04	1.717e−01	PROM1:15 AKR1D1:29 TGFB1:55 TNFSF10:72 RNASEL:151 PPARA:155
Neurodevelopmental Disorders	0.07436132	182	5.604e−04	1.718e−01	GABRB3:27 UBE3A:87 ANK2:118 L1CAM:188 EHMT1:252 SHANK2:312
Influenza	0.04889210	427	5.806e−04	1.726e−01	USP11:31 KRT5:36 CD9:54 TGFB1:55 TDRD7:57 TNFSF10:72
Bullous pemphigoid	0.14258187	48	6.369e−04	1.785e−01	ABCC11:16 DSG3:75 CD81:170 CD40:293 VCAM1:375 ICAM1:479
Trisomy	0.07773993	163	6.355e−04	1.785e−01	ABCC11:16 CHL1:45 IGF2:202 OAT:280 COL1A2:297 ETS2:316
Dermatitis, Allergic Contact	0.09946125	96	7.711e−04	2.045e−01	CNTN3:67 ACE:86 PPARA:155 CD80:244 MRC1:302 ETS2:316
Pemphigus Vulgaris	0.13386070	53	7.553e−04	2.045e−01	GJA8:59 DSG3:75 FCGRT:339 CD28:357 EGFR:373 PRKCA:470
Amyloid Neuropathies	−0.36591016	7	8.007e−04	2.057e−01	BRCA1:377 RBP4:405 APCS:570 GSN:1536 NGFR:1683 MPZ:1971
Hepatitis A	0.07080324	189	8.179e−04	2.057e−01	PROM1:15 AKR1D1:29 TGFB1:55 TNFSF10:72 RNASEL:151 IL33:172
Dementia	0.05123440	359	9.166e−04	2.193e−01	CLU:8 APOA1:23 TGFB1:55 KLK7:68 ACE:86 PRKN:89

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0008058 abnormal DNA repair	−0.15096312	39	1.140e−03	2.763e−01	TEX15:174 BRCA2:230 PMS2:732 OGG1:809 XRCC5:838 POLK:1163
MP0001968 abnormal touch/ nociception	0.15254247	33	2.480e−03	2.763e−01	TRPV1:154 SCN10A:585 SCN11A:599 GLRA3:680 GABBR2:709 PTGS2:1036
MP0002063 abnormal learning/memory/cond	0.04969449	351	1.830e−03	2.763e−01	CACNA1B:28 CHL1:45 UBE3A:87 IL4I1:136 TRPV1:154 SLC6A9:176
MP0003635 abnormal synaptic transmissio	0.04366575	421	2.903e−03	2.763e−01	ENTPD1:9 GABRB3:27 CACNA1B:28 UBE3A:87 KLK8:94 TRPV1:154
MP0010094 abnormal chromosome stability	−0.11653640	61	1.721e−03	2.763e−01	RAG1:184 BRCA2:230 MUS81:240 ATM:252 BRCA1:377 BUB1B:668
MP0002064 seizures	0.05206515	251	5.290e−03	4.196e−01	GABRB3:27 HRH2:32 UBE3A:87 KLK8:94 DGKD:106 SLC25A12:121
MP0002736 abnormal nociception after	0.15605402	22	1.140e−02	4.523e−01	SCN10A:585 SCN11A:599 GLRA3:680 PTGS2:1036 GRIA2:1725 TRPM8:1748
MP0002572 abnormal emotion/affect behav	0.04490281	289	1.014e−02	4.523e−01	GABRB3:27 CACNA1B:28 CHL1:45 NDST3:125 TRPV1:154 L1CAM:188
MP0008875 abnormal xenobiotic pharmacok	−0.18308712	16	1.131e−02	4.523e−01	ABCC3:124 SLC47A1:476 CYP1A2:497 NR1I3:662 AS3MT:743 AHR:1480
MP0001663 abnormal digestive system	0.08034312	86	1.047e−02	4.523e−01	TGFB1:55 IL6ST:206 UCHL1:372 SH3BP2:439 COL2A1:754 COL19A1:919
MP0004019 abnormal vitamin homeostasis	−0.11567323	41	1.060e−02	4.523e−01	RBP4:405 PLIN2:427 RPE65:1235 PTHLH:1288 LASP1:1336 RARG:1344
MP0009745 abnormal behavioral response	0.05579324	203	6.927e−03	4.523e−01	GABRB3:27 CACNA1B:28 KLK8:94 TRPV1:154 CD81:170 SLC6A9:176
MP0009384 cardiac valve regurgitation	−0.18059751	16	1.247e−02	4.568e−01	MUS81:240 ARSB:573 EFEMP2:1092 IDUA:1443 NFATC1:1541 GJA1:2311
MP0002837 dystrophic cardiac calcinosis	−0.22806687	9	1.789e−02	6.082e−01	DNM1L:274 SCG5:742 MYBPC3:795 SLC2A4:1010 GSTZ1:2840 DES:8709
MP0010771 integument phenotype	0.04889821	184	2.399e−02	6.433e−01	COL7A1:13 IL1RL2:53 DGKD:106 ST14:144 TRPV1:154 FBN1:213
MP0002067 abnormal sensory capabilities	0.04281002	257	2.040e−02	6.433e−01	GABRB3:27 CACNA1B:28 LTN1:44 CACNA1S:84 UBE3A:87 TRPV1:154
MP0001873 stomach inflammation	0.14323507	21	2.326e−02	6.433e−01	IL6ST:206 SH3BP2:439 IL18:1140 TOP3B:1146 IL4:1308 IFNG:1430
MP0004742 abnormal vestibular system	−0.10448937	39	2.433e−02	6.433e−01	PCDH15:48 HR:675 POU3F4:920 AP3D1:1060 NOX3:1104 BMP4:1482
MP0002735 abnormal chemical nociception	0.12188809	28	2.587e−02	6.481e−01	CACNA1B:28 TRPV1:154 HDC:643 PTGS2:1036 ADCY1:1042 TRPA1:1080
MP0002090 abnormal vision	0.09722561	42	2.972e−02	6.736e−01	TRPM1:50 ADORA2A:322 CLCN3:764 CDKN2A:887 NYX:900 POU4F2:924
MP0000358 abnormal cell content/	−0.05812953	120	2.921e−02	6.736e−01	DYNC2H1:10 CLIP2:96 DNM1L:274 BRCA1:377 FZR1:386 ARSB:573
MP0005365 abnormal bile salt	−0.12142992	26	3.240e−02	7.010e−01	ATP8B1:44 ABCC3:124 GPBAR1:498 HSD17B4:1251 ABCB11:1696 PEX5:2254
MP0002138 abnormal hepatobiliary system	−0.20137841	9	3.654e−02	7.100e−01	ABCB4:163 SOCS2:1719 MAP3K14:1832 HNF1B:2733 ARG1:2831 PKHD1:3270
MP0002882 abnormal neuron morphology	0.02243145	873	3.585e−02	7.100e−01	PDE6B:7 CLU:8 PROM1:15 GABRB3:27 LTN1:44 CNGB3:52
MP0002272 abnormal nervous system	0.04356169	195	3.864e−02	7.100e−01	GABRB3:27 CACNA1B:28 UBE3A:87 TRPV1:154 SLC6A9:176 CD86:181
MP0008469 abnormal protein level	0.03160927	384	3.878e−02	7.100e−01	TLR8:43 TGFB1:55 ACE:86 SLC34A3:109 TRAF2:171 CD86:181
MP0005334 abnormal fat pad	0.04465349	149	6.270e−02	7.382e−01	ACE:86 PYY:113 TRPV1:154 PPARA:155 RPTOR:408 VGF:415
MP0002733 abnormal thermal nociception	0.05586136	92	6.580e−02	7.382e−01	CACNA1B:28 TRPV1:154 L1CAM:188 ADORA2A:322 UCHL1:372 F2RL1:407
MP0005384 cellular phenotype	−0.02948182	419	4.484e−02	7.382e−01	DYNC2H1:10 ARID4B:22 CLIP2:96 ATG9A:136 CALR:137 AP2M1:154
MP0001727 abnormal embryo implantation	0.10999706	27	4.830e−02	7.382e−01	LIF:225 MRC1:302 CDC73:572 TYROBP:600 LPAR3:1018 PTGS2:1036
MP0005085 abnormal gallbladder physiolo	−0.10965961	25	5.813e−02	7.382e−01	ATP8B1:44 ABCB4:163 GPBAR1:498 ABCB11:1696 MUC1:1846 ACADL:2149
MP0010386 abnormal urinary bladder	−0.14423207	14	6.193e−02	7.382e−01	CHRNB2:188 CHRNA3:551 AGA:1555 PSAP:2540 CHRM3:2887 UPK2:2983
MP0000858 altered metastatic potential	0.07154434	58	6.052e−02	7.382e−01	CXCR2:189 CDH1:582 TOM1L2:642 CDKN2A:887 CASP1:896 EPHA2:1118
MP0001929 abnormal gametogenesis	−0.02768046	435	5.535e−02	7.382e−01	CREB3L4:39 ROS1:46 PPP1R3B:126 TEX15:174 TEKT2:180 PHYH:182
MP0002108 abnormal muscle morphology	0.06936440	64	5.607e−02	7.382e−01	CACNA1S:84 FBN1:213 COL1A2:297 LMX1B:363 EGFR:373 CASR:404
MP0003690 abnormal glial cell	−0.09152614	37	5.465e−02	7.382e−01	PHYH:182 ATM:252 KCNJ10:506 S1PR5:693 CP:1298 LGI4:1382
MP0003942 abnormal urinary system	−0.05978790	84	5.971e−02	7.382e−01	PPP3R1:33 PYGO2:352 ETV4:374 OSR1:566 BMP7:637 MKS1:692
MP0000579 abnormal nail morphology	0.12435130	19	6.091e−02	7.382e−01	COL7A1:13 L1CAM:188 LMX1B:363 EGFR:373 LRP4:406 GJB2:560
MP0008873 increased physiological sensi	0.04370835	151	6.669e−02	7.382e−01	TNFSF10:72 TRPV1:154 FLCN:173 IL6ST:206 RYR2:254 SMAD2:321
MP0003634 abnormal glial cell	0.03959203	203	5.534e−02	7.382e−01	PDE6B:7 LTN1:44 CD81:170 L1CAM:188 IL6ST:206 LIF:225

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
adrenal.gland	−0.055183650	108	4.869e−02	4.294e−01	CELA1:25 KLHL4:27 ABCC3:124 CYSLTR2:148 HTR2B:149 SLC47A1:476
Brodmann.area.9	0.050478208	180	2.056e−02	4.294e−01	GABRB3:27 GRM5:273 MAS1:314 UCHL1:372 ERC2:405 SCN1A:428
cerebral.cortex	0.056677161	107	4.386e−02	4.294e−01	LRTM2:287 MAS1:314 MGAT5B:485 PHF24:790 RHBDL3:951 MCHR1:1005
fallopian.tube	−0.084486398	52	3.550e−02	4.294e−01	ADGRG4:1 TCF23:368 HOXA7:825 HOXC5:893 CCDC17:1209 CFAP73:1269
leg.skin	0.040958421	217	3.961e−02	4.294e−01	GRIFIN:14 KRT5:36 TRPM1:50 IL1RL2:53 KLK7:68 DSG3:75
subcutaneous.adipose	−0.065442896	72	5.566e−02	4.294e−01	ITIH5:141 ACVR1C:176 PFKFB3:306 PLIN2:427 GPBAR1:498 LIPE:535
suprapubic.skin	0.047904407	193	2.298e−02	4.294e−01	COL7A1:13 CAMSAP3:33 KRT5:36 IL1RL2:53 DSG3:75 GSDMA:80
cerebellar.hemisphere	0.024681557	500	6.530e−02	4.408e−01	KCNK10:17 CHGB:82 NDST3:125 KIAA0895:157 MORN1:175 MYT1:179
Brodmann.area.24	0.088345959	31	8.911e−02	5.347e−01	FAM189A1:382 STUM:606 NCS1:916 SHANK1:1262 DRD5:1669 SLC6A1:1752
atrium.auricle	−0.037048126	90	2.264e−01	7.195e−01	ASB11:260 XIRP1:286 SMCO1:305 ALPK2:335 GHRH:371 ADPRHL1:392
cerebellum	0.020159752	374	1.887e−01	7.195e−01	CACNA1B:28 CHGB:82 ANK2:118 PCNX2:123 DGCR6:135 TAS2R3:160
ovary	−0.041357931	102	1.509e−01	7.195e−01	CCDC68:216 TCF23:368 UBXN8:369 CDH3:485 SLC8B1:502 SYTL4:646
skeletal.muscle	0.027762241	238	1.447e−01	7.195e−01	ANO5:62 CACNA1S:84 MYLK2:120 MYH7:137 CLTCL1:150 MYH2:164
spleen	−0.022957931	298	1.791e−01	7.195e−01	RASGRP2:24 FAM43A:69 SPNS3:115 RAG1:184 SAMD3:268 FCRL5:373
testis	−0.010765388	1375	2.143e−01	7.195e−01	INCA1:47 TTC21A:54 ZBED9:74 EME1:88 C1orf87:93 TMCO5A:97
transverse.colon	−0.028907309	149	2.265e−01	7.195e−01	ATP8B1:44 TRIM15:144 URAD:258 OMP:364 B3GNT6:538 MTMR11:572
vagina	0.040497576	90	1.861e−01	7.195e−01	KRT5:36 DSG3:75 PGLYRP4:99 TMPRSS11F:474 GJB2:560 LEXM:562
uterus	−0.042112673	60	2.605e−01	7.816e−01	TEX15:174 TCF23:368 CLEC4F:563 HOXD3:590 ENPP1:641 WT1:1196
C1.spinal.cord	0.022212185	182	3.056e−01	8.580e−01	PDE6B:7 SEC14L5:41 SLC6A9:176 PCSK6:243 KLHL32:260 UGT8:265
coronary.artery	−0.074180655	15	3.202e−01	8.580e−01	SUSD5:6 MYO18B:1471 SOST:1510 ST6GALNAC5:2395 GJA5:3172 CCN3:8709
liver	−0.015802234	327	3.337e−01	8.580e−01	CPN2:32 SERPINA7:127 ABCB4:163 DCXR:200 INHBC:255 OSGIN1:348
amygdala	−0.078537311	9	4.147e−01	9.555e−01	PCDH15:48 SLC38A8:1254 BCAN:1530 LHFPL3:8709 NEUROD2:8709 NEUROD6:8709
endocervix	0.028559917	72	4.036e−01	9.555e−01	FBLN1:292 ADRA2C:593 MASP1:1157 SNED1:1200 GABRE:1563 TRHDE:7257
hypothalamus	−0.028503277	66	4.247e−01	9.555e−01	PCDH15:48 LY6H:341 GHRH:371 PCDH19:406 SYNGR3:486 DIRAS3:983
aorta	−0.003468186	154	8.828e−01	9.611e−01	PCDH11X:2 SUSD5:6 C1QTNF8:38 COL8A2:493 PTGIS:550 OSR1:566
bladder	0.009928202	24	8.664e−01	9.611e−01	WFDC13:414 KRT13:1284 SLC14A1:1582 ACER2:7257 ARL14:7257 CLEC3A:7257
blood	−0.002304400	274	8.969e−01	9.611e−01	DHX34:151 JAK3:187 SAMD3:268 GPR84:349 SLA2:351 ADGRE1:419
breast	0.017750505	15	8.120e−01	9.611e−01	KRT5:36 LMX1B:363 ABCB5:7257 ADIPOQ:7257 CIDEC:7257 FABP4:7257
caudate.nucleus	0.042129459	26	4.576e−01	9.611e−01	MLC1:212 GABRG1:845 GPR37L1:1202 HS3ST5:7257 ANO3:7257 AQP4:7257
cortex.kidney	−0.005349120	114	8.444e−01	9.611e−01	TMEM252:210 SLC28A1:398 ESRRB:436 LRRC19:657 DCDC2:708 TMEM52B:859
EBV.lymphocyte	−0.005795781	668	6.214e−01	9.611e−01	STAMBPL1:12 IQGAP3:40 PLEKHG7:59 EME1:88 ERCC6L:158 BRCA2:230
ectocervix	−0.024062721	25	6.774e−01	9.611e−01	COL27A1:50 ADRA2A:941 PTHLH:1288 F10:1448 SOCS2:1719 GDF7:8709
esophagus.mucosa	0.008651484	220	6.617e−01	9.611e−01	CLCA4:26 KRT5:36 CAPN14:51 DSG3:75 PGLYRP4:99 IL36B:163
esophagus.muscularis.mucosa	−0.037168257	26	5.123e−01	9.611e−01	PRUNE2:7 COL4A6:420 RGMB:1034 CCDC69:1211 SYNM:2626 CHRM3:2887
greater.omentum	0.013019461	33	7.960e−01	9.611e−01	FFAR3:1448 MEST:1581 ADIPOQ:7257 CIDEC:7257 FABP4:7257 LEP:7257
hippocampus.proper	0.041443473	10	6.501e−01	9.611e−01	NEUROD2:7257 NEUROD6:7257 SLC17A7:7257 HRK:7257 OLIG2:7257 CABP7:7257
left.ventricle	−0.012389016	68	7.247e−01	9.611e−01	ASB11:260 XIRP1:286 SMCO1:305 ADPRHL1:392 TNNI3K:488 MB:628
lung	0.002311661	139	9.255e−01	9.611e−01	IL18R1:205 SUSD2:226 MRC1:302 TMEM100:473 ICAM1:479 RAMP3:486
minor.salivary.gland	−0.006940453	69	8.425e−01	9.611e−01	MMP7:130 EHF:159 MMP10:320 BPIFB2:347 MMP27:670 ITPRID1:761
nucleus.accumbens	−0.014754973	99	6.136e−01	9.611e−01	PCDH11X:2 DACH2:77 KCNA4:308 GPC5:524 HTR1D:653 OTOF:731